



वित्तीय सेवाएं विभाग
DEPARTMENT OF
FINANCIAL SERVICES



सेंट्रल बैंक ऑफ़ इंडिया
Central Bank of India

Cyber Security PSBs Hackathon Series – 2026. An Initiative by DFS & IBA via PSBs, hosted at MNNIT Allahabad, Prayagraj.

Note: Please do not modify the template. The proposal must be limited to a maximum of 3 pages. The completed document should be saved in PDF format and uploaded on the registration-cum-concept note form.

1. **Team Name:** DataDacoits

2. **Title of the Proposed Idea:**

CogniHaven- Continuous Behavioral Authentication for Digital Banking

3. **Problem Statement**

- Current digital banking systems rely on easily compromised static credentials and exhibit "blind trust" post-login, lacking continuous behavioral or contextual authentication. Furthermore, siloed perimeter defenses fail to correlate threat signals such as high risk IPs and unrecognized device fingerprints to block automated brute force attacks in real-time.

4. **Proposed Solution (Abstract):**

Our architecture tackles digital banking fraud through a progressive, three-layered defense system:

- **Layer 1: Perimeter Screening & Fraud Filter:** The system identifies and neutralizes automated threats before a login attempt is even processed. By analyzing attempt velocity and environmental context (i.e. geolocation, device being used, IPs, background processes) it applies deterministic rate-limiting to instantly block brute-force and credential-stuffing attacks.
- **Layer 2: Point-of-Entry Baseline Verification:** At login, the system evaluates physical interactions against the user's baseline. This generates an initial Risk Score that determines the adaptive response: frictionless entry, step-up MFA/OTP, or an immediate block.
- **Layer 3: Continuous Post-Login Monitoring:** Unlike traditional systems that stop checking once a user is inside, our engine runs continuously in the background. It monitors ongoing kinetic interactions (like mouse path curvature) and tracks high-risk session actions (e.g., attempting large transactions, changing a registered mobile number, or resetting a PIN). These actions dynamically adjust the live Risk Score. If the score spikes, the system instantly intervenes by prompting for security questions, requesting an OTP, or freezing the session.

Our solution bridges the gap between high security and low user friction by employing privacy-first data collection that captures strictly kinetic and timing metadata with zero PII. By leveraging unsupervised anomaly detection, the engine isolates complex fraud vectors without relying on rigid rule sets, guaranteeing a seamless experience for genuine users.



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5. PI Category:

Hybrid (Software & Cloud-Centric)

6. Technical Approach & Methodology:

Complete Idea (Data & Handling)

We replace static logins with a continuous behavioral engine using a background JS SDK that silently captures 5 telemetry categories: **Environment** (Canvas/WebGL fingerprinting, geo, VPN/TOR), **Keystrokes** (hold/flight times, error rate), **Mouse** (velocity, curvature, hesitation), **Session** (URL sequencing, paste detection), and **Transactions** (action velocity, Z-score variance). To ensure low latency, payloads bypass DB bottlenecks for instant in-memory Redis evaluation.

Technology Stack

- **Frontend:** React.js (mock portal & telemetry SDK).
- **Backend:** Python/FastAPI for async throughput and native ML integration.
- **DB & Cache:** PostgreSQL (historical baselines) paired tightly with Redis (ultra-fast in-memory cache for O(1) live Risk Score lookups).
- **AI Engine:** Scikit-learn using an Isolation Forest for unsupervised anomaly detection to isolate fraudulent feature vectors.
- **Deployment:** Docker for stable local/AWS demonstration.

Implementation Workflow

Security dynamically hardens across two adaptive phases.

- **Phase 1: Pre-Login Defense.**
 - Step 1 (Environment Check): Extracts hardware fingerprints, flags TOR/VPNs, and applies rate-limiting against bot brute-forcing.
 - Step 2 (Behavioral Biometric Validation): The Isolation Forest compares live keystroke dynamics and clipboard events against historical baselines to detect automated entry.
- **Phase 2: Post-Login Monitoring.**
 - Step 3 (Session Tracking): Monitors mouse curvature and navigation sequences to catch hijacked sessions/trojans.



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- Step 4 (Risk Engine): Aggregates signals into a real-time Redis Risk Score (0-100).
- Step 5 (Adaptive Response & XAI): At transaction, evaluates score: **0-30** (Frictionless), **31-65** (OTP triggered), **66-100** (Session will be Frozen). Flagged sessions trigger automated SOC reports detailing feature attribution (e.g., Blocked: Device mismatch +25, Anomalous typing +35).

7. Expected Outcome:

- Continuous Zero-Trust Security: Validates the user's identity throughout the entire session rather than just the entry gate thereby halting attacks that bypass traditional MFA.
- High Speed Decisioning: Delivers real-time risk scoring with a reduced latency, ensuring that high-velocity channels like UPI remain frictionless for genuine users.
- Improved Detection Accuracy: Utilizes unsupervised ML (Isolation Forest) to detect "unknown" fraud patterns and anomalies without requiring massive labeled fraud datasets
- Explainable Security: Provides reason for every blocked action (e.g., "Anomalous typing +35"), allowing bank SOC analysts to understand exactly why a session was flagged.
- Performance improvements:
 - By storing live Risk Scores in Redis, the system achieves O(1) lookups
 - Keeping it as a two layered check, initial checks + behavioural checks reduce the latency of the system making it a seamless experience.

8. Impact & Feasibility

- Potential impact: Protects banks and customers by neutralizing account takeovers and social engineering through invisible, continuous behavioral authentication that preserves a frictionless user experience.
- Scalability and real-world applicability:
 - Future employment of Apache Kafka and Apache Flink to process millions of concurrent events in real-time.
 - Future integration of RNNs/LSTMs to model user navigation as a sequence, identifying fraudsters who follow "unnatural" app pathways and Autoencoders to flag high reconstruction errors involved in behavioral patterns.
 - Scale into Graph Neural Networks (GNNs) to intercept organized fraud rings and mule networks by analyzing relational links between users and devices.
- Practical feasibility: The system captures only kinetic and timing metadata (zero PII), ensuring compliance with RBI and GDPR data minimization principles. The modular SDK is designed to be easily embedded across UPI, mobile apps, and internet banking portals with minimal changes to existing core banking logic.
- Challenges and limitations:
 - Behavioral Drift: Evolution in user habits over time can trigger false positives.
 - Cold Start: Lack of historical data for new users requires temporary fallback to traditional MFA
 - Network and Hardware Noise: Hardware and network variations require sophisticated feature normalization.